Artificial Intelligence Fraud Detection and Extractive Revenue Assurance in the Public Sector

A Case Study of the UK Department for Work and Pensions and a Proposal for AI-Assisted Extractive Revenue Assurance in Nigeria

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Introduction

Artificial intelligence (AI) has become one of the most consequential technologies in twenty-first-century government. The OECD's 2025 review of *Governing with Artificial Intelligence* catalogued more than 200 real-world deployments across 11 core public-sector functions. From welfare administration and tax collection to disaster response and judicial case management. The World Economic Forum reported that 65% of government entities surveyed planned to adopt AI tools by 2025, and reported returns on AI fraud-detection investments range from 10:1 to 15:1 in mature deployments. The US federal government alone loses an estimated $233 billion to $521 billion annually to fraud. Said losses that AI is increasingly being deployed to address.

Yet the OECD and the US Government Accountability Office both caution that AI in government is not self-implementing. Algorithmic bias, opaque methodology, fragmented IT infrastructure, and capacity gaps have repeatedly undermined deployments that looked promising on paper. The legitimate question for any public administrator is therefore not whether to adopt AI, but how to adopt it in ways that capture efficiency gains without eroding fairness, transparency, or public trust.

This report addresses that question in three steps. Section 2 examines the UK Department for Work and Pensions (DWP) machine learning system for detecting fraud and error in Universal Credit claims, one of the most thoroughly documented AI deployments in any government, with public data on both outcomes and limitations. Section 3 draws on those lessons to propose an AI-Assisted Extractive Revenue Assurance Platform for the Nigerian Extractive Industries Transparency Initiative, a context where underlying fiscal losses are roughly two orders of magnitude larger. Section 4 summarizes findings and recommendations.

Case Study Analysis: AI Fraud Detection at the UK Department for Work and Pensions

The UK Department for Work and Pensions administers one of Europe's largest social-protection programmes, distributing approximately £291 billion to 23 million claimants in the fiscal years of 2024 and 2025. That’s more than the UK spent on healthcare and roughly three times its defence budget. Annual fraud and error losses were estimated at £8 billion, concentrated in Universal Credit (UC) advance payments, self-employment claims, undeclared capital, and housing benefit. Rule-based screening could not keep pace with claim volume or sophisticated fraud rings, so the DWP committed approximately £70 million between 2022 and 2025 to advanced analytics, with a savings target of £1.6 billion by 2030.

In May 2022 the DWP placed into production a supervised machine learning risk-scoring model that screens every UC Advance application. The model learns from historical labelled outcomes, whether it be confirmed fraud or error versus confirmed legitimate) and consumes inputs including reported-income consistency, claim frequency patterns, and external data matches. There are three design choices that stand out about this model however: the model *triages* but does not auto-decide, any high-risk claims it makes go to caseworkers who retain full authority; pilots were sequenced across four discrete fraud typologies rather than one monolithic model; and companion generative-AI prototypes like "Aigent" and "a-cubed" support work coaches and policy drafting.

The UK National Audit Office reported in October 2025 that the model had saved approximately £4.4 million over three years in confirmed recoveries from the UC Advances pilot. A parallel global SAS survey of 1,100 public-sector fraud staff found that 57% cited "working more efficiently" as the primary AI benefit, ahead of fraud detection itself. The challenges, however, are equally well documented and inform the proposal that follows.

Documented Outcomes	Documented Challenges
£4.4M confirmed savings	Statistically significant bias across age, disability, marital status, nationality
Faster triage of high-risk claims; caseworkers focused on elevated-risk cases	~200,000 people wrongly investigated for housing benefit fraud

Workforce efficiency cited as top benefit by 57% of public-sector fraud staff	Only 1 of 9 protected characteristics fairness-assessed; demographic data gaps
Foundation laid to extend AI to higher-leakage benefits	Methodology redacted from public disclosure on "adversarial robustness" grounds
Human-in-the-loop architecture preserved	Fragmented IT systems and absence of cross-government data standards limit scale

Table 1 — UK DWP machine learning fraud detection: outcomes and challenges.

Five lessons generalize to any public-sector AI deployment: keep humans in the loop on consequential decisions; conduct fairness analysis *before* deployment; publish methodology, not just outcomes; invest in data quality before model sophistication; and target domains where both losses and weakness of pre-existing controls are large. These five lessons directly shape the proposal in Section 3.

Innovative Proposal - AI-Assisted Extractive Revenue Assurance for NEITI

Nigeria's extractive sector has historically generated up to 65% of government revenue and 85% of export earnings, yet has been chronically eroded by under-assessment, non-remittance, and crude theft. The Nigerian Extractive Industries Transparency Initiative, established under the NEITI Act of 2007, publishes annual reconciliation audits, performing transparency without delivering accountability

The fiscal scale of the problem far exceeds the DWP context, as Figure 1 makes visually clear:

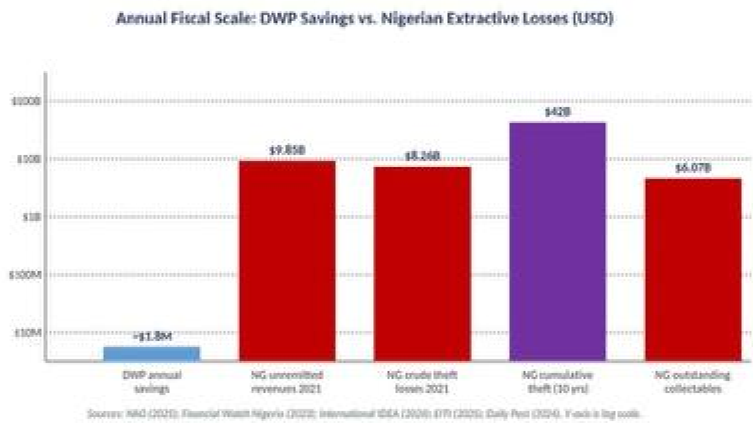

Figure 1 — DWP annual savings versus documented Nigerian extractive losses (USD, log scale).

I propose an AI-assisted continuous-assurance platform operating under NEITI's existing legal mandate, in partnership with the Nigerian Upstream Petroleum Regulatory Commission (NUPRC), the Federal Inland Revenue Service (FIRS), and the Central Bank of Nigeria (CBN). The platform shifts NEITI's role from retrospective annual audit to near-real-time revenue assurance, organized around four AI modules:

- Module 1 - Production-to-Royalty Reconciliation. Anomaly-detection models compare metered crude production from NUPRC telemetry, lifting records, and bill-of-lading data against royalty and tax payments recorded by FIRS and CBN, flagging mismatches at lease, joint-venture, and production-sharing-contract level.
- Module 2 - Beneficial Ownership Network Analysis. Graph neural networks ingest NEITI's beneficial-ownership register (established under the 2020 CAMA amendment) to surface ownership networks where the same controllers appear behind nominally unrelated awardees and counterparties. Globally, network-analysis adoption is forecast to expand from 32% to 87%.
- Module 3 - Document and Contract Intelligence. Large-language-model document analysis processes production-sharing contracts, marginal-field award letters, and pre-export financing agreements to extract fiscal terms and obligations from unstructured text.
- Module 4 - Crude-Theft Detection. Computer-vision models applied to satellite imagery (Sentinel-1 SAR, Sentinel-2 optical) detect bunkering vessels and illegal refining sites; time-series models reconcile metered-production drops against expected curves. The 2022–2023 NEITI audit documented crude losses of 36.7 million barrels, 22.46% of total metered production.

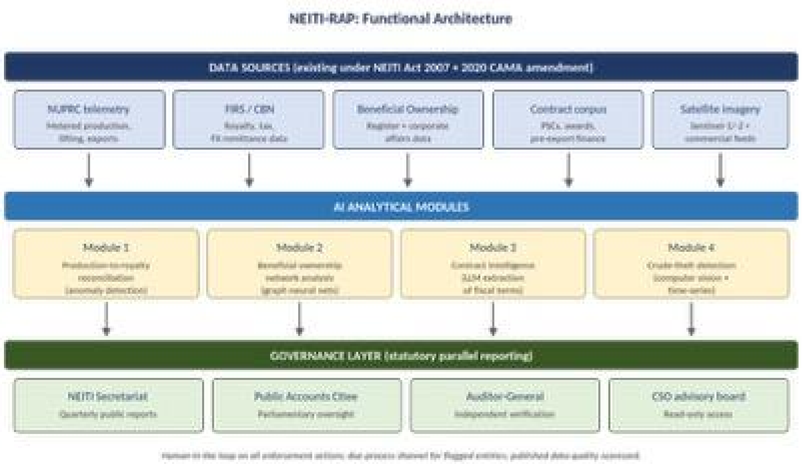

Figure 2 — NEITI-RAP three-layer architecture: data sources, AI modules, and statutory parallel-reporting governance layer.

The Nigerian extractive context inverts most of the DWP risk profile, making it unusually well-suited to AI assistance:

Dimension	UK DWP (welfare fraud)	Proposed NEITI-RAP (extractive revenue)
Targets	Vulnerable individuals; risk of dehumanization	Corporate entities and transactions
Data type	Sparse, demographic, bias-prone	Structured numerical: production, prices, royalties
Annual fiscal leverage	£4.4M saved on £291B disbursed (~0.0015%)	$8–10B documented annual losses → 10% recovery = $600M–$1B
Bias risk	**High** — protected characteristics implicated	**Medium** — firm-level procedural fairness required
Legal authority	Expanded surveillance powers contested by civil society	Existing NEITI Act 2007 + 2020 CAMA amendment
Worst-case harm	Wrongful payment suspension; rent arrears; food insecurity	Enforcement action against a firm with rebuttal process

Table 2 — Why NEITI-RAP inverts the DWP risk profile.

A conservative assumption that the platform recovers 10% of currently documented unremitted revenues, well below the 10:1 to 15:1 returns McKinsey reports for AI fraud-detection investments in comparable settings, implies recoveries of $600 million to $1 billion in the first three years, roughly two orders of magnitude larger than the DWP's confirmed savings. Beyond the fiscal outcome, the platform shortens NEITI's reporting cycle from annual to quarterly, restoring the political-economy mechanism that EITI was designed to create: timely disclosure that civil society and parliament can act on.

Challenge	Why it matters	Mitigation
Inter-agency data sharing	Only 7% of public-sector fraud teams collaborate cross-agency; NUPRC, FIRS, CBN sit under different ministries	Begin with NEITI–NUPRC MoU pilot on Module 1 for one fiscal year and one basin; demonstrate value before broader access
Political capture	NEITI itself critiqued as "hybrid organisation" vulnerable to the interests it monitors	Statutory parallel reporting to NEITI, Public Accounts Committee, and Auditor-General, eliminating points of suppression.
False positives against firms	Can damage legitimate operators, deter FDI, create legal-challenge grounds. A DWP cautionary template	Mandatory pre-deployment fairness analysis; published model-risk framework; rebuttal channel before public disclosure
Data quality	Production-metering gaps; historical operator-vs-regulator volume discrepancies	Publish a data-quality scorecard alongside every output; quantify reconciled vs unverified inputs. Serves as an inverse of DWP's redaction approach

Capacity and skills	60% of public-sector IT cite limited AI skills as top challenge	Structure as 36-month capacity-transfer programme with embedded NEITI/NUPRC staff progressively assuming ownership
Cybersecurity	Nearly all public-sector fraud teams report being targeted by AI-powered attacks	NIST AI Risk Management Framework standards; signed audit logs; segregation from any payment-execution authority

Table 3 — Anticipated challenges and design mitigations.

Conclusion

This report examined AI in the public sector through one well-documented case study and developed an innovative proposal informed by that case. Three findings emerge.

First, AI in the public sector delivers real but context-dependent value. The DWP machine learning model saved £4.4 million over three years and improved caseworker triage, but on a £291 billion programme this is a vanishingly small share of total disbursements. The lesson is that AI's leverage depends heavily on the underlying loss rate and the strength of pre-existing controls.

Second, fairness and transparency are design constraints, not afterthoughts. The DWP's own fairness analysis revealed statistically significant disparities across age, disability, marital status, and nationality, and approximately 200,000 people were wrongly investigated for housing benefit fraud. The Department's reluctance to disclose its methodology has eroded public trust in ways that may outlast the technical pilot.

Third, the highest-leverage AI applications may lie outside the welfare-fraud paradigm. The Nigerian extractive-revenue context inverts the DWP risk profile on every dimension that matters: corporate rather than individual targets, structured rather than personal data, two-orders-of-magnitude larger losses, and existing legal-institutional infrastructure already in place. NEITI-RAP could conservatively recover $600 million to $1 billion in the first three years while shortening NEITI's reporting cycle from annual to quarterly.

As for recommendations, there are four practical suggestions. One, target domains where documented losses substantially exceed expected implementation cost. Two, preserve human-in-the-loop authority on consequential decisions. Three, publish methodology and fairness analysis *before* deployment. Four, structure deployments as capacity-transfer programmes so institutional ownership accumulates within government rather than within external vendors.

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